Size-dependent demography, disturbance, and transition dynamics of *Acropora palmata* across the Caribbean

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Abstract

Restoration of endangered corals depends on colonies surviving, growing, and retaining live tissue after outplanting. For the elkhorn coral *Acropora palmata*, evidence on these processes is scattered among natural-reef monitoring programs and restoration experiments conducted under different environmental conditions. We synthesized longitudinal observations from 18 Caribbean studies to evaluate size-dependent survival, growth, tissue loss, and movement among five size classes. Larger colonies survived more often, but proportional growth declined with size and surviving colonies frequently lost tissue. In a recruitment-free transition model that included fragmentation, persistence of the largest colonies contributed most to population growth; the deterministic λ was 0.955, indicating incomplete replacement within this subsystem. Survival varied widely among studies, and differences in colony source and study setting prevented a ranking of outplanting practices. Simulated mortality matching the 2023 Florida heatwave further reduced effective λ to 0.853 even when events recurred only once every 50 years. These findings connect outplanting success to the survival and subsequent size of the original cohort, rather than survival alone. They also distinguish evidence for maintaining large living colonies from the reproductive and habitat recovery that restoration programs ultimately seek.

Keywords: Caribbean coral reefs; demographic synthesis; elkhorn coral; meta-analysis; population dynamics; restoration

Introduction

Restoring endangered reef-building corals requires knowing which colonies survive after planting and whether they grow enough to replace lost living cover. Caribbean acroporid declines have reduced coral cover and the habitat once provided by *Acropora palmata* (Aronson and Precht 2001; Gardner et al. 2003). White-band disease, storms, predation, and chronic tissue loss continue to alter the abundance and condition of remaining colonies (Gladfelter 1982; Rogers and Muller 2012; Williams and Miller 2012). Restoration programs propagate and outplant *A. palmata*, but rebuilding populations requires more than adding colonies: planted corals must survive, grow, and retain live tissue (Edmunds and Riegl 2020). Population rehabilitation also differs from recovery of ecosystem structure and function (Croquer et al. 2026). Evidence on colony demography comes from both restoration experiments and monitoring of natural reefs, scattered among regions and disturbance histories. Bringing those observations together could identify recurring demographic constraints and show which restoration decisions still lack comparable evidence.

Colony size is one restoration choice for which demographic evidence is directly relevant. Practitioners also choose propagation material, nursery treatment, attachment, and planting sites; these choices differ in their effects on early survival and later performance (Forrester et al. 2012; Williams et al. 2023; Ramos Romero et al. 2025). Large colonies can survive, reproduce sexually, and produce fragments, whereas small corals can grow rapidly in proportional terms but experience high mortality (Highsmith 1982; Hughes 1984; Hughes and Jackson 1985; Lirman 2000; Madin et al. 2014). This contrast makes size at outplanting a potential trade-off between retaining colonies and increasing their area. Survival alone does not resolve that trade-off. Storm damage, disease, and predation can remove live tissue from survivors and move them into smaller classes (Lirman and Fong 1997; Madin et al. 2020). Partial mortality can also affect reproductive output: *A. palmata* colonies with unhealed lesions produced smaller eggs than apparently healthy colonies (Piñón-González and Banaszak 2018). A synthesis of size-dependent survival, growth, and tissue loss can therefore evaluate whether the evidence supports a common size-related pattern, rather than treating survival as a complete measure of recovery.

Demographic rates also depend on the conditions experienced during monitoring. Storm frequency and intensity alter population trajectories in stage-structured simulations (Lirman 2003), while field observations and larval experiments document effects of disturbance at different life stages (Randall and Szmant 2009; Rogers and Muller 2012). Disease outbreaks can cluster within local populations (Lentz et al. 2011). Consequently, differences among studies may reflect exposure history as well as colony size. Excluding disturbed intervals would remove much of the environment in which restoration operates, but pooling them without retaining exposure information could conceal the conditions associated with a rate. The 2023 marine heatwave, which caused functional extinction of *A. palmata* in Florida (Manzello et al. 2025), illustrates the consequence. Survival measured before the event omits its mortality pulse; projections using those earlier rates alone cannot represent the loss caused by its recurrence.

Synthesis is useful only if it preserves these differences among studies. Nursery outplants, colonies re-cemented after fragmentation, direct outplants, and natural fragments differ in initial size, handling, and environment (Forrester et al. 2012; Banister et al. 2024; Ramos Romero et al. 2025). Early survival and later performance can respond to different processes

(Williams et al. 2008; Williams et al. 2023). A pooled estimate for "restoration fragments" may therefore describe the mix of sources sampled rather than the effect of a particular practice. Combining studies can quantify the range of observed survival and reveal how source composition affects that estimate. It cannot rank interventions when treatment, site, and colony size vary together. That distinction defines what the existing evidence can contribute to husbandry and outplanting decisions.

Demographic models connect these observations to the maintenance and replacement of colonies. Models for *Acropora cervicornis* in Puerto Rico have evaluated how fragment size, fragmentation, and survival affect population trajectories (Mercado-Molina et al. 2015), while the principal size-structured model for *A. palmata* used tagged colonies in the northern Florida Keys (Vardi et al. 2012). We combined individual records from seven studies with summaries from eleven additional studies to address three linked questions: how survival, growth, and tissue loss varied with colony size; how much annual survival varied among studies and colony sources; and which transitions contributed most to colony replacement in a model without sexual recruitment. We retained disturbance histories to describe the conditions underlying the observations, then applied Florida heatwave mortality as a separate projection scenario. This approach evaluates demographic patterns that can inform monitoring and outplant-size trials without attributing differences among unrelated studies to restoration treatments.

Materials and methods

Literature search and study selection

From June through December 2025, we searched repositories maintained by NOAA and the US Geological Survey, structured Google Scholar results, backward and forward citations from focal demographic studies, and data shared by restoration practitioners. In March 2026, we expanded the search through PubMed, Web of Science, Semantic Scholar/Elicit, targeted citation searches, and the project literature library. Search terms combined *Acropora palmata* or elkhorn coral with survival, mortality, growth, recruitment, restoration, and long-term monitoring. Additional individual colony records were obtained by direct data sharing in April 2026. We assembled a public data and analysis repository containing the search strings, screening decisions, and study-level exclusions (https://github.com/stier-lab/rse-2025-detmer-apalmata-rse-parameters; Online Resource 1, Table S1).

We included studies that reported longitudinal observations of identified *A. palmata* colonies or fragments, survival or areal growth over a defined interval, and sufficient information to annualize survival or estimate a size-dependent vital rate. We excluded cross-sectional condition surveys, studies that reported only branch extension, laboratory micro-fragment experiments smaller than 1 cm^2, and overlapping analyses of the same tagged-colony datasets. Titles and abstracts were screened first, followed by full-text evaluation. A second reviewer independently evaluated a subset of expanded-search candidates; disagreements underwent a third review. Across all sources, we identified approximately 2,500 records, assessed 101 distinct study or data sources at full text or repository level, and excluded 83 with documented reasons; the most common reasons were a different species, overlap with an included dataset, and cross-sectional designs. The screening ledger contained 104 rows; we consolidated one duplicate Gladfelter record and three Vardi entries representing the same underlying study.

When a source supplied an eligible parameter but another parameter was excluded, we counted the study once as included.

The final dataset comprised 18 studies. Seven supplied individual colony records: NOAA long-term monitoring of tagged colonies in the Florida Keys, Curaçao, and Navassa; Neely et al. (2022); Pausch et al. (2018); Kuffner et al. (2020); a USGS outplant experiment in the US Virgin Islands; FUNDEMAR nursery fragments in the Dominican Republic; and Mendoza Quiroz et al. (2023). Eleven supplied study-level summaries (Rogers et al. 1982; Bruckner and Bruckner 2001; Ortiz Prosper 2005; Garrison and Ward 2008; Williams and Miller 2010; Vardi 2011; Rogers and Muller 2012; Forrester et al. 2014; Maurer et al. 2022; Rosales et al. 2024; Ramos Romero et al. 2025). Seventeen of these studies, contributing 22 study-region effects, entered the survival synthesis. Mendoza Quiroz et al. (2023) contributed only to the size-dependent analyses (Online Resource 1, Table S1). Online Resource 1 summarizes the verified selection counts; early identification totals were approximate and could not support a fully enumerated PRISMA flow.

Data extraction and standardization

We classified records into two tiers. Tier 1 studies supplied individual colony identifiers, repeated sizes, and survival fates; these records supported analyses of survival, growth, tissue loss, and size-class transitions. Tier 2 studies supplied study-level survival proportions, sample sizes, and, where available, mean colony size. The standardized survival file contained 7,882 interval records. Deduplication by study, colony identifier, survey year, location, and source interval identifier where available removed 66 records; removing 36 records with missing survival or missing or non-positive starting size left 7,780 prepared records. The growth file contained 6,318 records; removing 19 duplicates left 6,299 prepared records. These counts describe colony intervals, which can include repeated observations of the same colony.

We expressed colony size in cm² using each source's area measurement or area index. Where perpendicular dimensions and live-tissue proportion were available, live area was length × width × proportion live tissue, following the convention used in previous *A. palmata* size-structured models (Vardi et al. 2012). Neely et al. (2022) instead supplied a live-area index, $LAI = [(L + W + H)/3]^2 \times p_{live}$, where L, W, and H are colony dimensions and p_{live} is the fraction of living tissue. We retained that index as a size proxy; it is not a photo-traced planar area. For diameter-only records, we estimated a circular footprint; single linear measures required study-informed width-to-length ratios. We refer to these comparable but nonidentical measurements collectively as area. In the Neely data, a literal "live" entry established survival without a measured LAI; it was never coded as zero area. We retained such fates in the source inventory, while size-dependent analyses required a measured starting size and growth required sizes at both endpoints. Distinct survey rounds, including two rounds in 2012, retained their source interval identifiers. Missing survey dates were not imputed (Online Resource 1, section 1).

We annualized non-annual survival as $S_{annual} = S_{observed}^{1/t}$, where t is the monitoring interval in years. This transformation assumes a constant hazard within each interval and therefore does not represent seasonal mortality peaks. Mortality definitions also differed among studies, from loss of both tissue and skeleton in NOAA monitoring to loss of at least half of live tissue in Kuffner et al. (2020). We retained each study's definition rather than imposing a common one,

so part of the among-study variation reflects measurement rather than biology. The Neely index can also combine several tissue isolates on an older skeleton; equal index values need not indicate equal age, intact colony size, or reproductive condition. Known clonality at Dry Tortugas and Marker 3, and suspected clonality elsewhere, further means that colony counts cannot be interpreted as numbers of independent genotypes (Grober-Dunsmore et al. 2007).

We assigned colonies to five size classes (SCs): SC1, 0 < area ≤ 10 cm²; SC2, 10 < area ≤ 100 cm²; SC3, 100 < area ≤ 900 cm²; SC4, 900 < area ≤ 4,000 cm²; and SC5, area > 4,000 cm². We retained the 100, 900, and 4,000 cm² boundaries of the published *A. palmata* model to preserve comparability (Vardi et al. 2012), and divided its smallest class at 10 cm² to distinguish the smallest colonies. Vardi et al. linked the larger boundaries to branch development and increasing clonal and sexual reproductive potential. Our classes describe live-area ranges, not verified life stages or reproductive states: a small colony may be a recruit, a fragment, or a larger colony that lost tissue. We called any negative annual change in a surviving colony's measured live area tissue loss. Following the size-class transitions in Vardi et al. (2012), we used shrinkage for movement into a smaller size class; retrogression is the equivalent matrix term used in the figures. Tissue loss could occur without shrinkage across a class boundary. Stasis meant survival in the same size class, despite possible gains or losses of tissue.

We calculated relative growth rate as $RGR = G / A_0$, where G is annual areal change (cm² yr^{-1}) and A_0 is initial colony area (cm²), giving RGR in yr^{-1}. We used live-tissue area and live-tissue change where available, with total-colony measurements as fallbacks. Negative RGR indicates a decrease in measured area; positive RGR indicates an increase.

Disturbance and restoration classification

We included disturbed intervals in the primary analyses. To describe their context, we matched monitoring regions and calendar-year windows to a curated timeline of storms, disease, bleaching, cold events, and chronic pressures. Regional entries applied only to their named regions; Caribbean-wide entries applied throughout the dataset. The window ran from the starting survey year through the final calendar year implied by the interval duration, with a one-year matching window when duration was missing or shorter than one year. Overlap therefore indicated a potential regional exposure, not measured exposure or diagnosed mortality in an individual colony. The classification distinguished no curated overlap, context-only pressure, and overlap with an event flagged for baseline exclusion. That last flag, rather than the timeline's duration label, determined removal in the acute-event sensitivity analysis: for example, the prolonged 2005 thermal anomaly was flagged for exclusion, whereas the 2014–2017 bleaching regime remained contextual. Direct coral disease, chronic disease pressure, and indirect community effects were recorded separately in the chronology, but could overlap within an interval. The supplement gives worked examples and checks of the interval assignments (Online Resource 1, Table S2). We tested size-by-category associations in natural colonies; those observational comparisons did not estimate event-specific mortality.

For the colony-source comparison, we distinguished naturally occurring fragments, nursery-held material, and reef outplants. Pausch et al. (2018) supplied nursery outplants; Kuffner et al. (2020) supplied nursery outplants re-cemented after fragmentation. FUNDEMAR observations came from fragments held on nursery tables, not reef outplants. The USGS reef outplants had insufficient source information to establish direct wild-to-reef transplantation. We separated

sexually propagated Mendoza Quiroz et al. (2023) corals held at Puerto Morelos from those planted at Cuevones and Picudas. Only NOAA records explicitly flagged as fragments entered the natural-fragment category, not the full NOAA monitoring dataset. Because that flag was assigned at the interval endpoint among living records, survival of this selected fragment subset was not estimable as cohort survival. We displayed its growth where measured but withheld its survival estimate. These definitions distinguish life stage, source, and observation setting without attributing among-study differences to a restoration practice (Online Resource 1, Fig. S5).

Size-dependent vital rates

To evaluate size-related patterns, we fitted binomial generalized additive models (GAMs) for survival and Gaussian GAMs for relative growth, using log-transformed colony size as the predictor. The survival curve used natural colonies after excluding subannual intervals; the relative-growth curve used all colony sources, with the live-area-preferred and total-area fallback rule described above. Points and fitted curves in each panel came from the same model population. These descriptive curves pooled observations without adjustment for study. We evaluated sensitivity to study differences with mixed-effect models including a study-level random intercept; additional diagnostic models included site, year, or colony effects where supported by the data. Following Detmer et al. (2025), we first compared each GAM with a linear model. The threshold gate required an improvement of at least 2 AIC units (AICc for samples smaller than 40); all primary curves used AIC. Derivatives then located changes in slope or curvature. A second-derivative minimum identifies strongest downward curvature, not a mathematical inflection, which would require a change in the sign of curvature. These features of fitted curves were not tested minimum sizes for outplanting. We checked model fit and repeated the analyses with each study excluded in turn to assess their dependence on study composition.

Survival synthesis

To summarize annual survival and its variation among studies, we aggregated annualized survival proportions into study-region effects. For Tier 1 data, we estimated survival within groups sharing study, region, population type, and interval duration, annualized those proportions, and averaged them within effects using the number of interval records as weights. This synthesis used 7,610 of the prepared records: 52 Mendoza Quiroz et al. (2023) records were outside the survival-synthesis study set, 24 Neely et al. (2022) records lacked survey dates needed for duration, and 94 endpoint-selected NOAA fragment records were excluded from the natural-colony NOAA effects (NCEI accession 0 142 175; (Williams and Miller 2016)). Tier 2 effects contributed a further denominator of 1,431, giving 9,041 across 22 effects. Within the hand-extracted Tier 2 summaries, we used the maximum initial sample size within each effect rather than summing repeated census denominators, and weighted annualized survival by initial sample size. The total denominator therefore combines Tier 1 colony-interval records and Tier 2 effect-level sample sizes; it is not a count of unique colonies. The broad restoration-origin group includes nursery-held material as well as reef outplants; it is not a pooled estimate of post-outplant survival alone.

We transformed survival proportions to log odds, which compare survival with mortality on an unbounded scale. When an effect contained no survivors or no deaths, we added 0.5 to both

counts so that the log odds and sampling variance remained finite. The primary random-effects model used restricted maximum likelihood (REML), with study and within-study effect structure and *t*-based inference; an independent-effect model with Knapp–Hartung inference was a sensitivity analysis (Knapp and Hartung 2003; Viechtbauer 2010). We report pooled survival, 95% confidence intervals, prediction intervals, and heterogeneity statistics. We evaluated population type, colony size, year, and region as exploratory moderators because the number of effects was limited. We also repeated the synthesis after excluding individual effects, changing classification assumptions, and restricting studies by methodological quality. We assessed small-study asymmetry with Egger-type regression tests on both models. A trim-and-fill sensitivity analysis added hypothetical effects to make the independent-effect funnel plot more symmetric and recalculated pooled survival. These hypothetical effects were not additional observations. Funnel-plot methods were designed for comparative effect sizes, so we treated them as diagnostics of asymmetry among single-proportion effects rather than direct tests of publication bias.

For the separate colony-source comparison, we displayed study–region summaries and record-weighted subtype means without uncertainty bars in either panel. Growth used only records with measured initial live area and live-area change, without total-area substitution. The small number of studies per subtype could not support comparable between-program uncertainty for survival and growth; variation among plotted studies remained descriptive. The endpoint-selected NOAA fragment subset was excluded from cohort-survival comparisons.

Transition model

To connect individual rates to colony replacement, we constructed a five-class Lefkovitch matrix describing annual changes in the abundance of natural colonies. The matrix included survival, growth, shrinkage to smaller classes, and fragment production; sexual recruitment was fixed at zero. Its dominant eigenvalue (λ) therefore described annual growth of this recruitment-free model, not growth of the full population. Four studies contributed survival rates and two contributed growth rates. NOAA monitoring (Williams and Miller 2016) supplied most SC5 survival observations, and Vardi (2011) supplied the only fragmentation estimates. We calculated matrix-cell elasticities: the proportional change in λ expected from a small proportional change in each cell, with other cells held constant. To propagate sampling uncertainty, we resampled studies, then observations within sampled studies. We retained 2,000 valid bootstrap replicates; 151 imputed survival for at least one missing size class. The bootstrap did not propagate survival–growth covariance. We also excluded each of the four contributing studies in turn, refitting the same survival estimator and growth transitions. Excluding Vardi (2011) also removed its fragmentation estimates; missing survival classes used the full-data estimates.

For transitions conditional on survival, we required measured initial live area or the source live-area index and its annual change, without substituting total-colony area. We excluded records whose observed live endpoint was zero, so that deaths entered the survival estimates rather than the growth transitions. We restricted natural-colony records to 0.5–1.5-year intervals, removed biologically impossible growth and the upper 1% of absolute live-area changes, and retained 4,698 records. We projected one-year area as initial live area plus annual live-area change, assigning projected areas below 0.1 cm^2 to 0.1 cm^2. This linear annualization

approximated a common interval, rather than directly observing every colony after exactly one year; a positive observed endpoint could project below zero and receive the floor.

To test whether independent survival and growth resampling affected uncertainty, we compared two additional 2,000-replicate bootstraps. Both retained two draws from the studies shared by survival and growth and two from survival-only studies; the coordinated version used the same shared-study draws for survival and growth. This comparison tested study-level coordination while holding the study mix fixed, but could not estimate temporal survival–growth covariance. We also replaced missing-class survival with a draw from its full-data bootstrap distribution, compared this with point imputation and discarding incomplete replicates, and removed the survival and fragmentation contributions of Vardi (2011) separately (Online Resource 1, Table S3).

Heatwave scenarios

We applied the 2023 Florida heatwave mortality as an additional episodic survival pulse to the pooled reference matrix. The monitoring records extend through 2024; this matrix was not fitted as a pre-heatwave counterfactual, and the scenarios do not reconstruct the historical event. We first used the *A. palmata* dose-response thresholds reported by Manzello et al. (2025) (ED50 = 7.8 degree-heating weeks [DHW], ED95 = 17.6 DHW) to define 8-, 12-, and 18-DHW sensitivity scenarios. We then added an observed-2023 Florida endpoint: 97.8–100% mortality, represented by a 0–2.2% heatwave-year survival multiplier. The deterministic projection used the midpoint of that range; because the published study reports bounds rather than a colony-level sampling distribution, Monte Carlo projections sampled uniformly across the reported range at each prescribed event. Each of 500 simulations per scenario retained the full-data matrix structure, scaled by one sampled bootstrap lambda for all 50 years; these simulations propagated parameter uncertainty, not environmental variation in annual transition rates. Heatwave pulses were applied after the annual matrix transition to all size classes, consistent with the reported absence of size refuge during the Florida event. We report the geometric mean of annual abundance ratios as effective lambda under recurrence intervals of 5, 10, 20, and 50 years, and the fraction of simulations in which total abundance fell below 10% of its initial value at year 50 (quasi-extinction). We prescribed these recurrence intervals to compare frequent and rare events; we did not estimate future return periods. The projections describe the recruitment-free subsystem under additional Florida-like mortality and cannot estimate Caribbean-wide thermal mortality or full population viability.

To test sensitivity to matrix structure and initial size composition, we projected all 2,000 complete bootstrap matrices and compared these with matrices scaled by the corresponding bootstrap lambdas. Each matrix remained fixed for 50 years, and heatwave mortality draws were identical across paired comparisons. The original starting population contained 100, 50, 30, 20, and 10 colonies in SC1–SC5; alternative starting proportions were 70, 15, 8, 5, and 2% (small-colony-heavy), their exact reverse of 2, 5, 8, 15, and 70% (large-colony-heavy), and the full-data matrix's stable size distribution. These compositions were sensitivity assumptions, not estimates of Caribbean population structure (Online Resource 1, Table S4).

Reproducibility

We conducted analyses in R 4.5.2 under the project's locked computational environment; the central workflow uses mgcv for generalized additive models, metafor for meta-analysis, and lme4

for mixed-effect diagnostic models. The public repository contains the data-processing and analysis scripts, figures, and tables, with links from each main-text result to its generating script and output (Data availability statement below). A versioned archival release will be deposited upon acceptance.

Results

Evidence spanned the Caribbean but remained geographically uneven

Demographic records covered 13 Caribbean regions, but the Florida Keys supplied 6,656 observations (85.6% of the 7,780 prepared individual-level survival records), more than all other regions combined (Fig. 1). Several regions contributed records from only a subset of size classes. Records of the largest colonies were concentrated in Florida, Curaçao, and Navassa, leaving their survival and transitions poorly documented elsewhere.

Survival increased with size while proportional growth declined

Survival generally increased with colony size, although size explained only 8.2% of deviance in individual fate among the 5,711 fitted natural-colony records (Fig. 2a). The survival GAM fit better than a linear model (ΔAIC = 9.2), with strongest downward curvature near 7,498 cm²; the estimated colony size at this curvature minimum depended strongly on which study was retained. Survival therefore increased gradually rather than identifying a stable cutoff between low- and high-survival colonies. Relative growth declined with size across 4,997 records from all colony sources, with strongest downward curvature near 35 cm² (2.5th–97.5th percentiles across leave-one-study-out fits: 33.3–38.7 cm²). The positive-growth model did not pass the nonlinearity gate, so we did not report a corresponding threshold (Online Resource 1, Fig. S1). Neither fitted curvature nor the pooled survival pattern established an experimentally tested outplant size. Absolute-growth variability increased strongly with size, while proportional changes were less variable in large colonies.

Study-level survival estimates by size class showed that the size effect was concentrated in the largest classes (Fig. 2c). Pooled survival was nearly flat from SC1 to SC3 (77.9%, 75.5%, and 77.2%) and then rose to 91.3% in SC4 and 94.9% in SC5. Among-study spread was widest in SC1–SC3, where study context, disturbance history, and population type also varied most strongly. Colony size alone did not account for the wide variation among studies.

Annual survival varied widely among Caribbean studies

The expanded synthesis estimated pooled annual survival at 78.9% (95% CI: 70.7–85.2%; 95% prediction interval: 38.0–95.8%) across 17 studies and 22 study-region effects (Fig. 3a). Effects calculated from individual colony records (Tier 1) and published study summaries (Tier 2) were both annualized before aggregation. Heterogeneity was high (I^2 = 97.6%; τ^2 = 0.71, of which 0.28 was between studies and 0.43 within studies), so the pooled value summarizes the assembled evidence rather than a Caribbean-wide constant. Excluding any one parent study changed the primary estimate by at most 1.6 percentage points; excluding individual effects changed the independent-effect estimate by at most 2.5 points. Restricting the independent-effect synthesis to the 13 studies with the highest methodological-quality scores changed it by less than 1 point. Evidence for small-study asymmetry was mixed: the Egger-type test was not significant for the independent-effect model (p = 0.71) but was significant for the three-level

model (p = 0.001). Adding five hypothetical effects to make the independent-effect funnel plot more symmetric lowered its estimate from 80.3% to 74.8%; those effects were not observed studies.

Pooled survival was 85.7% for natural-colony effects (95% CI 73.6–92.8%, k = 10) and 74.5% for restoration-origin effects (95% CI 63.9–82.9%, k = 12). The natural-colony estimate was 11.1 percentage points higher, but uncertainty was too large to distinguish the groups in the moderator test (p = 0.16). With 22 effects, the estimated difference needed for 80% power was about 20 percentage points, so the observed difference could remain unresolved even if real. The analysis therefore supported neither a reliable ranking nor a claim of equal survival. Regional variation remained substantial within population types (Fig. 3b). Even where the two types overlapped in colony size, study identity and sampling context could not be separated from population origin (Online Resource 1, Fig. S2).

Surviving natural colonies often lost tissue

Surviving natural colonies lost live tissue in 39.3% of 4,698 near-annual growth records from two studies (Online Resource 1, Fig. S3). Tissue-loss frequency increased from 22.5% in SC1 to 44.2% in SC5. In the same records, annualized downward-class probabilities were 5.5% in SC2, 14.5% in SC3, 17.4% in SC4, and 11.4% in SC5. Movement into a larger class occurred in 47.5% of SC1 records (n = 40), 27.0% of SC2 (n = 637), 13.9% of SC3 (n = 1,375), and 10.0% of SC4 (n = 1,269). SC5 had no higher modeled class, although colonies could gain area within it. These transitions were conditional on survival and projected to one year from observed intervals. Tissue loss was more frequent than movement into a smaller class because colonies could lose area without crossing a class boundary.

Size–survival associations differed across exposure categories

Of 1,075 monitoring intervals, 95.1% overlapped a documented disturbance or chronic pressure (Online Resource 1, Table S2). Size–survival relationships differed among the three exposure categories in natural colonies (global likelihood-ratio test, p = 0.001; Online Resource 1, Fig. S4); the size-by-category interaction was also supported for positive growth (p = 0.005). The cell percentages in Fig. S4 are descriptive, not pairwise tests of SC4 or SC5 survival across states. In particular, the no-curated-overlap group represented a restricted set of studies and years and included only one SC1 survival record, with no corresponding growth records. Removing baseline-exclusion intervals changed both source coverage and pooled survival in the individual-record sensitivity analysis. These comparisons did not isolate mortality caused by the events or demonstrate a survival benefit of disturbance.

Restoration survival varied among colony-source subtypes

The subtype inventory contained 1,257 survival records from six studies, of which 1,163 could describe cohort survival after withholding 94 endpoint-selected NOAA fragments (Online Resource 1, Fig. S5). Record-weighted survival was 57.4% for Pausch nursery outplants (n = 968), 81.1% for re-cemented Kuffner outplants (n = 53), and 65.2% for USGS outplants of unspecified source (n = 46). FUNDEMAR nursery-held fragments had survival of 86.4% (n = 44). Mendoza Quiroz sexual cohorts had no observed deaths in either the nursery subset (n = 17) or reef-outplant subset (n = 35). These were descriptive means from different sites and cohorts,

not estimates of source effects. Only four studies supplied the paired live-area measurements required for the growth panel (n = 857); missing growth estimates were not treated as zero.

Largest-class stasis dominated conditional transition dynamics

The recruitment-free transition matrix estimated a deterministic lambda of 0.955 (95% bootstrap interval: 0.916–0.998; Fig. 4). In 98.0% of hierarchical-bootstrap replicates, lambda was below one. This resampling fraction characterizes uncertainty in the compiled transition rates; it is not a probability of real-world population decline. Stasis dominated the transition matrix, and SC5-to-SC5 stasis accounted for 51.4% of matrix-cell elasticity (95% bootstrap interval: 38.8–64.5%). Leave-one-study-out estimates of lambda ranged from 0.888 (excluding Vardi (2011), including its fragmentation estimates) to 0.966 (excluding NOAA monitoring (Williams and Miller 2016)). Excluding Neely et al. (2022) gave lambda of 0.956. These exclusions used the same survival estimator and transition filters as the primary matrix. Removing fragmentation altogether lowered lambda to 0.890, so asexual fragment production, estimated from a single study, contributed about 0.065 to the deterministic growth rate. Within this conditional, recruitment-free subsystem, persistence of the largest colonies had more influence on lambda than marginal changes in any single early transition.

Most bootstrap estimates remained below replacement under alternative resampling choices: 98.9% of lambda estimates were below one with coordinated shared-study draws (95% interval: 0.924–0.993), compared with 97.7% under independent resampling with the same study mix (Online Resource 1, Table S3). Lambda was below one in 97.4% of the 2,000 replicates under both point and distribution-based missing-class imputation; their upper interval bound was 1.0004, just above replacement. Removing Vardi's (2011) survival data alone lowered lambda to 0.953, whereas removing its fragmentation estimates alone lowered it to 0.890. The combined exclusion effect therefore arose mainly from fragment production.

Florida-scale heatwaves reduced effective lambda

Adding heatwave mortality reduced projected abundance without changing the underlying pooled reference matrix. Under the observed 2023 Florida endpoint, effective lambda was 0.357, 0.579, 0.772, and 0.853 when a Florida-like event recurred every 5, 10, 20, and 50 years, respectively (Online Resource 1, Fig. S6). In these primary simulations, every observed-endpoint scenario was below the quasi-extinction threshold at year 50 in all 500 simulations. Without heatwaves, 81.0% of the 500 primary simulations were below the year-50 threshold, and effective lambda was 0.938. This finite-horizon rate differed from the deterministic eigenvalue of 0.955 because it included bootstrap uncertainty and transient changes in size structure. The 18-DHW dose-response sensitivity scenario was less severe than the observed endpoint because its fitted mortality was 95.5%, rather than the 97.8–100% mortality reported from the Florida Keys and Dry Tortugas (Manzello et al. 2025). In these recruitment-free projections, recurrence of the observed Florida mortality drove abundance below the quasi-extinction threshold even at the longest return interval. The scenarios do not forecast recovery across the Caribbean.

Complete-matrix uncertainty gave similar baseline projections for the original size composition: effective lambda was 0.938 and 78.3% of simulations fell below 10% of initial abundance at year 50 (Online Resource 1, Table S4). In the paired comparison using all 2,000 bootstrap draws, 79.6% of lambda-only simulations were below the year-50 threshold. Initial

size composition had a larger effect: with full matrices and no heatwaves, the fraction below the threshold ranged from 19.5% for large-colony-heavy populations to 90.0% for small-colony-heavy populations. Under observed Florida mortality, at least 99.9% of simulations were below the threshold across all four starting compositions and prescribed return intervals.

Discussion

Restoration programs need to know whether adding colonies will rebuild living coral cover and sustain it after planting. Larger *A. palmata* colonies generally survived more often, but survivors frequently lost tissue, and persistence in the largest class contributed most to our recruitment-free model. Earlier work established the importance of large colonies and the consequences of fragmentation (Lirman 2000; Vardi et al. 2012). Our synthesis quantifies tissue loss among survivors and connects size-dependent survival, growth, and shrinkage in a common model with uncertainty. A small outplant that grows rapidly but dies and a large outplant that survives while losing tissue contribute differently to the planted cohort. Trials that follow survival and retained live area across the original cohort can test which planting treatments improve these outcomes (Table 1). Reproductive and habitat recovery require additional observations (Croquer et al. 2026).

Colony size connects survival to persistence

The demographic contribution of a surviving colony depends on the size it retains. Size-structured theory captures this distinction by separating survival from growth and movement to smaller classes (Hughes 1984; Hughes and Jackson 1985). In our matrix, large-colony stasis had the greatest elasticity, agreeing with the northern Florida Keys model of Vardi et al. (2012) and with Lirman's (2000) observations of persistence, reproduction, and fragment production in large *A. palmata* colonies. Agreement with Vardi is partly expected because Florida Keys monitoring also dominates our largest-size-class data. Survival did not ensure retention of colony size: tissue loss occurred in 44.2% of SC5 growth records, and 11.4% of surviving colonies moved out of SC5 in the annualized transitions. Disease and corallivore stress on large colonies in St. John (Grober-Dunsmore et al. 2006) provide a further reason to distinguish whole-colony mortality from partial loss of live tissue. Live-area loss need not imply an equivalent loss of physical habitat, however, because the standing skeleton may remain after tissue dies. Our area records do not measure skeletal persistence or three-dimensional complexity.

First fertility and widespread reproductive maturity are different demographic milestones. In Panama, Soong and Lang (1992) recorded a fertile *A. palmata* colony measuring 16 × 8 cm, but estimated a puberty size of 1,600 cm² from size-ranked observations of fertility. Their area estimate summed the living surface area of branches and the colony base, unlike the area measures and live-area index pooled here. These observations support increasing reproductive potential with size, but cannot assign fertility directly to our classes. Neither growth into SC5 nor retrogression out of it establishes a change in reproductive status.

For a program that has selected *A. palmata* and a planting site, colony size remains a separate choice from nursery treatment and attachment. Relative growth declined with size and changed curvature near 35 cm², whereas survival generally increased over a much larger size range. The growth threshold therefore does not imply that planting above 35 cm² improves

growth, or that planting below it improves cohort survival. Nor does the unstable survival curvature establish a minimum planting size. Published size comparisons also differ: larger initial fragments survived more often in long-term monitoring (Forrester et al. 2014), whereas smaller fragments produced more tissue without a detected survival difference in a single-genotype experiment (Pausch et al. 2018). Neither the observational comparison nor the single-genotype experiment establishes a size that performs best across genotypes and sites. Comparing initial sizes within shared genotypes and sites would test whether greater survival offsets slower proportional growth.

Survival estimates depend on study setting and disturbance history

An annual survival estimate is useful for restoration planning only in relation to the setting it represents. The 38.0–95.8% prediction interval shows why the pooled mean cannot serve as a single expected survival rate for a new study setting. Grouping intervals by documented disturbance did not remove those differences: the small comparison group without documented exposure came from a restricted set of studies and years, rather than matched colonies at the same sites. Its lower observed survival in some classes therefore provides no evidence that acute disturbance protected large colonies. Selecting a survival rate requires information on site, survey period, colony source, and mortality events, beyond a disturbed or undisturbed label.

Storm-driven simulations (Lirman 2003) and the 2023 Florida observations (Manzello et al. 2025) address mortality pulses that interrupt between-event demography, rather than differences among pooled exposure categories. Manzello et al. reported 97.8–100% mortality of *A. palmata* in the Florida Keys and Dry Tortugas. Applying that mortality to our matrix reduced effective λ even when events recurred only once every 50 years. The scenario does not estimate future heatwave frequency. It shows that the compiled monitored rates, including fragmentation, did not compensate for these mortality pulses in our recruitment-free projections. Outplant evaluations conducted over short intervals can characterize establishment; they cannot establish persistence through disturbances they did not observe.

Selecting stock for heat resistance requires evidence that an early advantage persists through later demographic stages. Karp et al. (2025) found higher acute heat tolerance in *A. palmata* hosting *Durusdinium* than in colonies hosting *Symbiodinium*. Miller et al. (2024) documented lower bleaching among sexually propagated juveniles in a multi-species Caribbean comparison, but also severe effects in an older, reproductive *A. palmata* cohort. These observations motivate trials across host genotypes or families, symbiont associations, and sites, with follow-up through recovery and reproductive maturity (Table 1). Acute assays and juvenile bleaching scores alone cannot establish lifetime resilience. Our pooled size relationships likewise cannot identify heat-resistant stock or isolate effects of sexual origin from age and rearing history.

Geographic coverage and differences in measurement also limit the transfer of pooled rates to a particular restoration site. Florida Keys studies supplied 85.6% of prepared individual-level survival records. Colonies outside Florida, especially the largest colonies, were poorly represented; studies also differed in mortality definitions and monitoring intervals. In the Neely records, LAI combined colony dimensions with live-tissue proportion, sometimes across several tissue isolates, and was not interchangeable with age or intact planar area. Site-level

responses, including the collapse at Marker 3 during 2014 bleaching, and known or suspected clonality also limited the independent information carried by colony counts. Site, year, and colony diagnostic models addressed repeated observations, but neither the pooled curves nor study-level bootstrap resolved genotype-specific or site-specific disturbance responses. Most matrix bootstrap estimates remained below replacement under alternative resampling, although one sensitivity interval extended just above one. The separate Vardi exclusions showed that fragment production contributed more to the change in λ than removal of its survival observations. Applying these pooled rates locally therefore requires checking the measurement convention, source composition, and disturbance history against the target site.

Scaling from local colonies to connected populations would require fecundity, fertilization, larval dispersal, settlement, and early survival. A spatially explicit metapopulation model could test whether recruitment from surviving source reefs offsets local losses. Our recruitment-free matrix cannot estimate that rescue or regional extinction risk.

Safeguarding genetic material is a separate conservation outcome from maintaining colonies on reefs. Muller et al. (2026) describe how restoration infrastructure, rescue efforts, and living gene banks preserved genetic richness during the 2023 Florida losses. Inventories of genotypes across reef sites and backup facilities could identify holdings without redundancy or with shared exposure (Table 1). Estimating the benefit of diversification or rescue would require spatial-risk and intervention data absent from our model. Genotype retention and rescue survival therefore require their own evaluation rather than being inferred from reef-colony λ.

Retaining colonies and rebuilding populations require different actions

Care of existing colonies can limit losses while planting adds colonies. Persistence in the largest live-area class strongly influenced the modeled growth rate, giving a specific reason to test whether targeted care reduces whole-colony mortality or tissue loss in that class. These comparisons should include care of wild colonies, not only maintenance of outplants. Snail removal reduced snail abundance and feeding scars on *A. palmata* in a field experiment (Williams et al. 2014), providing evidence for one action where predation is documented. Pooled tissue loss cannot identify which colonies need predator control, disease intervention, or stabilization. A Cuban experiment also found differences in fragment survival and growth among benthic organisms and reef crests (Ramos Romero et al. 2026); these effects do not justify treating every neighbor as harmful. Replicated comparisons of targeted care, planting, their combination, and unchanged practice could test which actions retain more living area (Table 1). Elasticity motivates that comparison but does not measure an intervention's effect.

Planted cohorts contribute through both survival and subsequent size change. Following every original outplant, with repeat area measurements and dead colonies counted as zero, would show whether rapid growth among survivors compensates for losses elsewhere in the cohort (Madin et al. 2025). Trials could compare clonal sizes within shared genotypes and sites, then report retained cohort area and the fraction reaching a target live-area class. Source and observation setting must remain separate: nursery-held fragments are not reef outplants, and naturally occurring fragments are not deliberate plantings. The endpoint-selected NOAA fragment records could describe growth among surviving fragments but could not estimate their cohort survival. Re-cemented colonies in Kuffner et al. (2020) provide a distinct planted-colony example; testing reattachment itself would require comparable broken colonies with

and without that intervention. First-year reef survival also begins with a different denominator from survival through larval rearing and nursery production (Chamberland et al. 2015; Mendoza Quiroz et al. 2023). Reporting those stages separately would make biological comparisons interpretable. Costs and production time would be additional requirements for comparing cost-effectiveness, not for estimating demographic efficacy.

Neither retention nor a larger planted cohort establishes sexual replacement. Sexually propagated *A. palmata* have reached fertility and spawning (Mendoza Quiroz et al. 2023), and mixed-genotype planting groups can support fertilization (Baums et al. 2019). But more clonal colonies need not mean more genotypes, and spawning need not produce surviving recruits. Tests of planting arrangements should therefore follow identified genotypes through fertility, fertilization, settlement, and recruit survival, using parentage where feasible (Table 1). Croquer et al. (2026) define effective return as outplants surviving to become reproductive; their subsequent contribution to population replacement requires recruitment observations absent from our matrix. Recovery of habitat structure likewise requires measurements beyond live area.

The projections make the demographic shortfall explicit. Even with fragmentation, the compiled rates did not replace colonies within the recruitment-free subsystem, and Florida-scale heatwave mortality drove all 500 primary observed-endpoint simulations below 10% of initial abundance at year 50. These are severe conditional outcomes, not species-extinction probabilities: sexual recruitment and immigration were absent, and future heatwave frequency was not estimated. They give urgency to testing whether care and planting improve retention through disturbances, whether heat resistance persists to reproductive maturity, and whether surviving genotypes produce recruits. Safeguarding genetic material can preserve options during that work (Muller et al. 2026), but it is not itself recovery of a self-sustaining reef population.

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Statements and Declarations

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Competing interests

The authors declare no competing interests.

Ethics approval and permits

Not applicable. This study synthesized existing observations and published study summaries and involved no new human-participant research, animal use, or field collection requiring ethical approval or permits.

Consent to participate

Not applicable.

Consent to publish

Not applicable.

Data availability

The data and code underlying this study are maintained at https://github.com/stier-lab/rse-2025-detmer-apalmata-rse-parameters. A versioned archival release will be deposited upon acceptance.

Author contributions

A.C.S. conceived the study and led manuscript development. A.R.D. led data curation and analyses, made the figures, and contributed to manuscript review. K.L.N. contributed Florida Keys data, data interpretation, and manuscript review. T.V. contributed the transition-matrix framework, fragmentation estimates, and interpretation. J.F.S., D.B., A.C., and R.I.S.-B. contributed domain expertise, data resources, and interpretation. S.D.G.-G. led the FUNDEMAR fragment-nursery work and contributed colony-size and area data. M.F.V. contributed FUNDEMAR data validation.

Online resources

Online Resource 1 (ESM_1.pdf). Concise supporting analyses for the *Acropora palmata* demographic synthesis: four supporting tables, including study coverage, disturbance overlap, matrix resampling, and projection sensitivities, and six supplementary figures. Extended diagnostics and source outputs remain available in the public repository.

Tables

Table 1. Restoration priorities require different measures of recovery. Proposed comparisons and actions for *Acropora palmata*, with measurements needed to evaluate their benefits. These are unranked research priorities, not intervention benefits estimated by the synthesis. Biological comparisons require matched follow-up intervals and appropriate controls; a separate cost-effectiveness analysis would also require costs.

Priority	Proposed comparison or action	Measurements needed
Retain established colonies	Compare targeted care, additional planting, their combination, and unchanged practice in replicated site blocks. Snail removal provides one tested local action (Williams et al. 2014).	Change from baseline in total live area, colony survival, and genotype retention. Elasticity alone does not estimate treatment benefit.
Compare planting choices	Compare clonal sizes within shared genotypes and sites; retain age, family, and rearing history for sexual cohorts (Pausch et al. 2018; Mendoza Quiroz et al. 2023).	Total retained cohort live area, counting dead outplants as zero; fraction reaching a target size; losses before planting. Separate nursery-held material, reef outplants, and natural fragments.
Rebuild reproductive neighborhoods	Compare spatial arrangements of identified genotypes at comparable initial sizes and source composition (Baums et al. 2019).	Survival to fertility, spawning overlap, fertilization, settlement, and recruit survival; parentage where feasible. Colony counts alone cannot measure recruitment.
Follow heat resilience through maturity	Compare hosts or families and symbiont associations across sites, distinguishing manipulations from existing associations (Miller et al. 2024; Karp et al. 2025).	Measured heat exposure, age, rearing history, tissue recovery, growth, and reproduction through maturity. Juvenile resistance need not persist in adults.
Prepare for shared losses	Map genotype holdings and exposure across reef sites and backup facilities; establish rescue responsibilities, capacity, and response triggers (Muller et al. 2026).	Redundancy, losses shared among holdings, rescue survival, and genotype retention. Spatial benefits cannot be estimated from the current matrix.
Diagnose local losses	Separate attachment failure, disappearance, death, predation, disease signs, and overgrowth; test cause-specific treatments against controls (Ramos Romero et al. 2026).	Colony survival and retained tissue by diagnosed cause and life stage. Fragment-stage effects cannot establish benefits for larvae or early recruits.

Figure legends

Fig. 1 Study locations and size-class coverage for the *Acropora palmata* demographic synthesis. (a) Study locations across the Caribbean. Circles identify individual-level (Tier 1) studies and triangles summary-level (Tier 2) studies; regions with both are shown with both symbols. Symbol area is proportional to the number of observations, and labels give regional totals. Locations are regional centroids. (b) Availability of individual-level observations by size class and region; cell labels are sample sizes, and darker cells indicate more observations. Empty cells indicate no available observations. Size classes are SC1, 0–10 cm^2; SC2, 10–100 cm^2; SC3, 100–900 cm^2; SC4, 900–4,000 cm^2; and SC5, >4,000 cm^2. Each class includes its upper boundary and excludes its lower boundary; zero-area records were excluded.

Fig. 2 Size-dependent demographic rates for *Acropora palmata*. (a) Survival probability against initial area for natural colonies, excluding subannual intervals. The line and shaded band are an unadjusted binomial generalized additive model (GAM) and its 95% confidence interval; points are binned observed proportions. Upper and lower rugs identify survival and mortality observations. (b) Relative growth rate (yr^{-1}) for all colony sources, using live-area change where available and total-area change otherwise. The line and band are the unadjusted Gaussian GAM estimate and 95% confidence interval; pale points are individual observations, gold points are binned medians, and the dashed vertical line and band show the curvature minimum and the 2.5th–97.5th percentile range across leave-one-study-out fits. Area includes the Neely live-area index, not exclusively planar measurements. Points, bins, and rugs in each panel use that panel's fitted records. (c) Study-level annual survival by size class. Colour indicates population type, shape identifies data tier, and point area is proportional to sample size; diamonds and error bars are pooled estimates and 95% confidence intervals. *k* gives the number of contributing effects.

Fig. 3 Caribbean-wide survival synthesis for *Acropora palmata*. (a) Random-effects annual-survival estimates from 17 studies contributing 22 effects. Points and horizontal lines are study estimates and 95% confidence intervals; point area is proportional to random-effects weight. Circles represent individual-level data and triangles summary-level data. Coloured diamonds are population-type pooled estimates, the dark diamond is the overall pooled estimate, the dashed line marks that estimate, and the shaded interval is its 95% prediction interval. (b) Regional study estimates and 95% confidence intervals. Colour identifies population type, point area is proportional to sample size, and dark diamonds identify regional pooled estimates where at least two effects were available. The dashed line marks overall pooled annual survival.

Fig. 4 Conditional transition dynamics for *Acropora palmata*. (a) Five-class, recruitment-free Lefkovitch transition matrix. Cells give annual matrix entries from size class at time t to size class at time $t + 1$; colour encodes entry magnitude. Entries combine survival and size-class movement with fragment production in the SC5 column, which consequently sums to more than one (Vardi 2011). (b) Elasticity decomposition by source size class. Stacked bars show stasis, growth, and retrogression; orange points show fragmentation elasticity, which is displayed separately because it is included in the retrogression entries rather than an additional transition class. (c) Distribution of lambda from 2,000 hierarchical-bootstrap replicates. Gray bars denote replicates with lambda < 1 and green bars replicates with lambda at least 1; the distribution characterizes uncertainty in the compiled transition rates, not the probability of real-world decline. The dashed line is the deterministic estimate and the solid line marks replacement. (d) Leave-one-study-out lambda estimates. The dashed and dotted lines mark the full-data estimate and replacement, respectively; orange identifies exclusion of the NOAA survey.

Figures

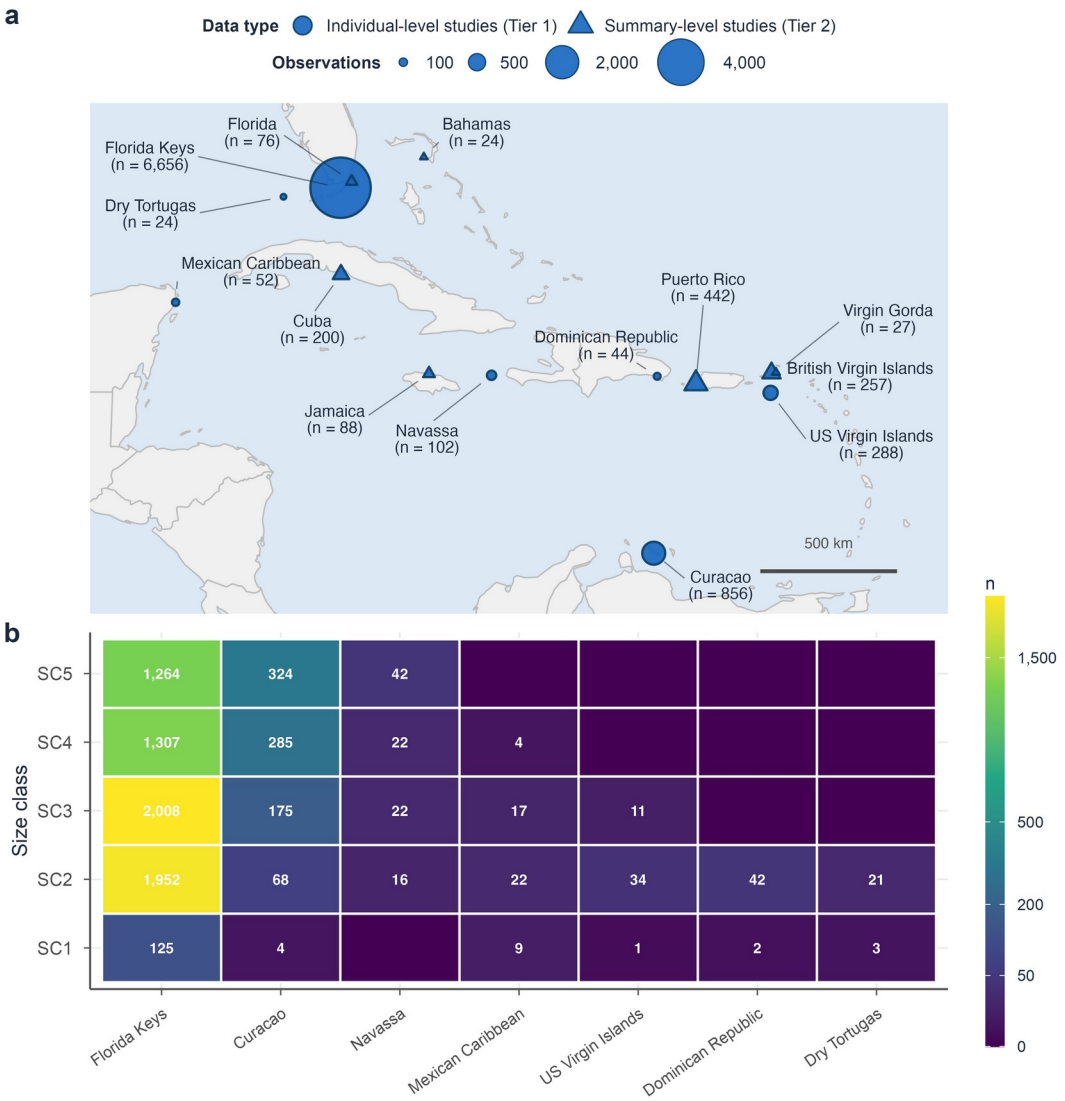


Figure 1. Study locations and size-class coverage for the *Acropora palmata* demographic synthesis.

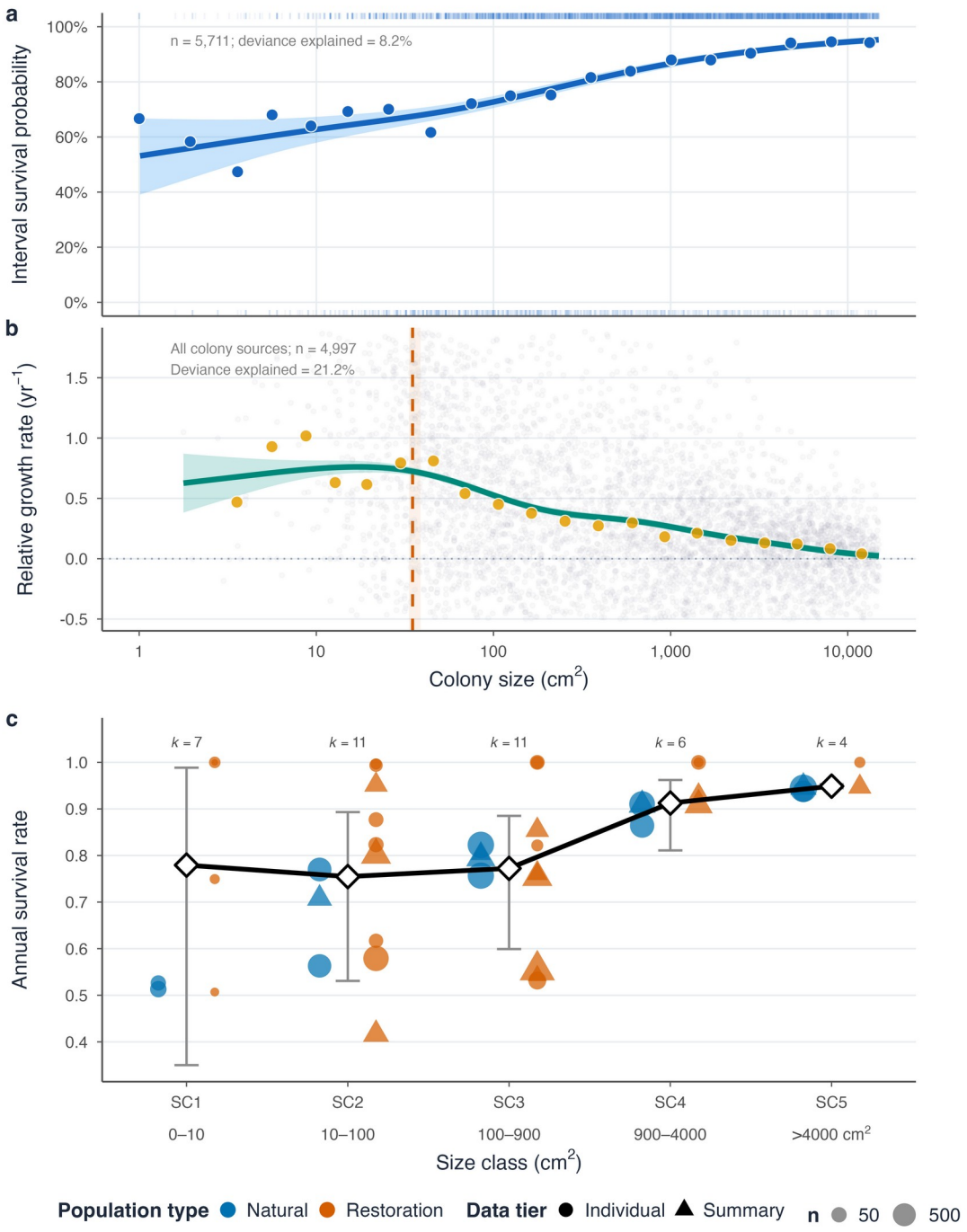


Figure 2. Size-dependent survival and growth rates for *Acropora palmata*.

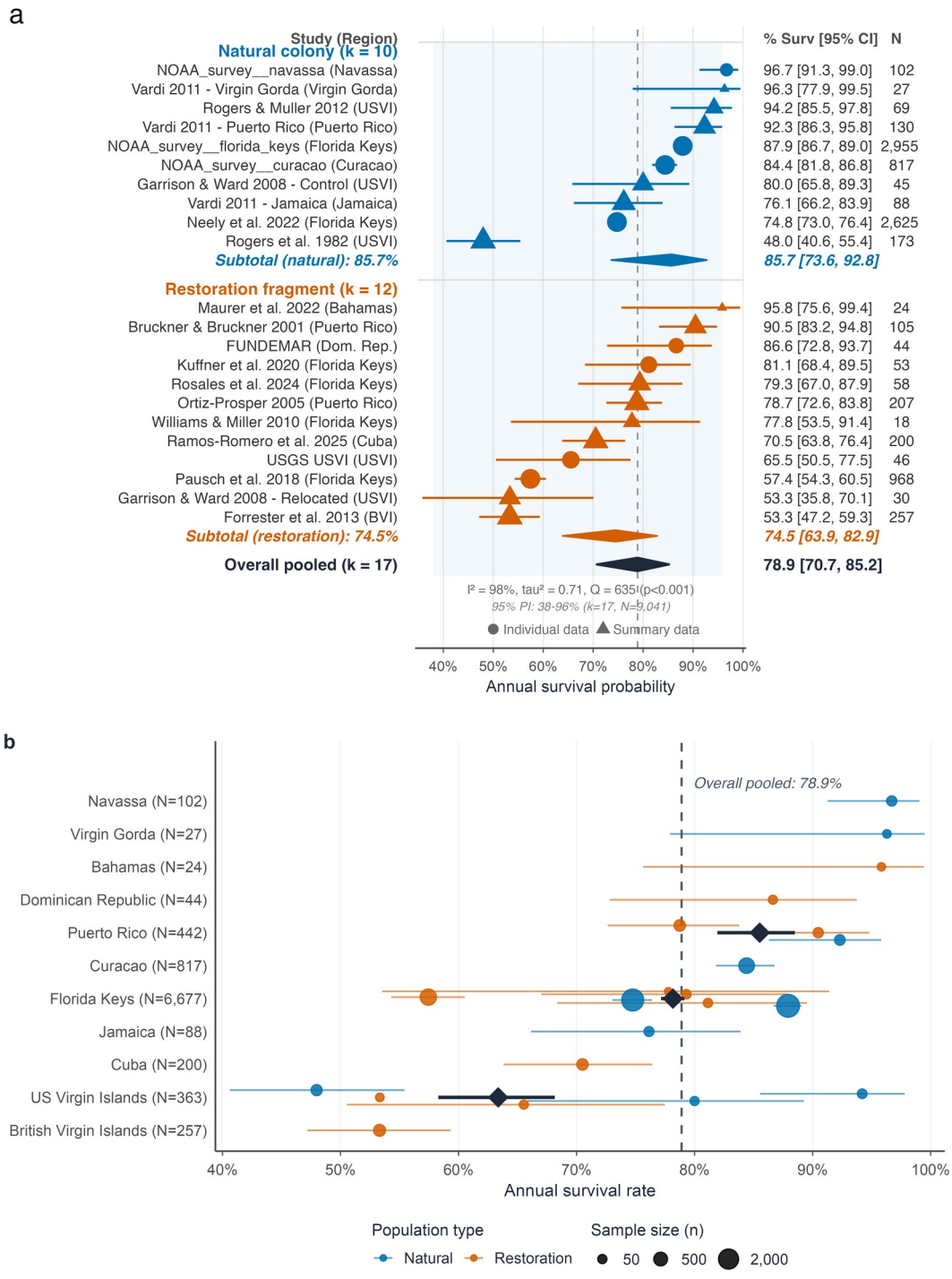


Figure 3. Caribbean-wide annual-survival synthesis for *Acropora palmata*.

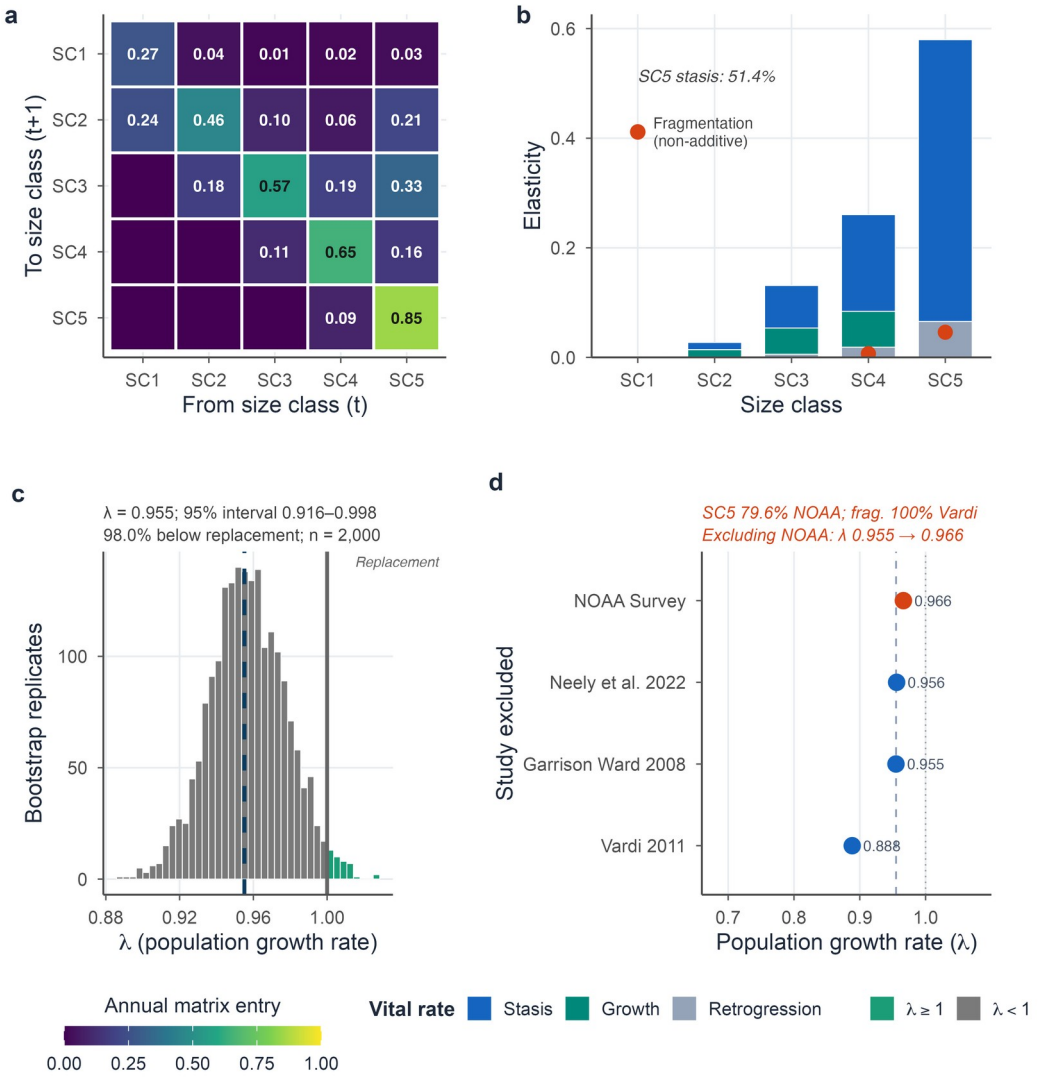


Figure 4. Conditional recruitment-free transition dynamics for *Acropora palmata*.

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